Summing up: it appears that, if we have good reason to believe the multiverse hypothesis, this would be bad news for the fine-tuning argument. But it also seems that the fact that our universe is life-supporting is not itself evidence for the multiverse hypothesis. So the key remaining question is: do we have any good reason to believe in the multiverse?

This is a question very much in dispute — though the dispute is as much among physicists as philosophers. Some physicists think that there is physical evidence in favor of the multiverse hypothesis. Others think that the very idea of physical evidence about universes distinct from our own makes little sense.

Here — as in other cases — we have an example in which philosophical reasoning and scientific theory are intertwined.

What seems clear is that if (1) there is just one universe and (2) current thinking about the fundamental constants in physics is on the right track, then the fine-tuning version of the design argument is a powerful argument for the existence of a designer of the universe.

fine-
tuning of the
universe

Bayes' theorem

the argument

objections

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Enough about probability (for now). Let's turn now to some results from contemporary physics which will be important to the argument which follows.

A first question: what does the theory provided by physics include?

‘‘The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.’’ (Hawthorne & Isaacs, ‘‘Fine-tuning fine-tuning’’)

These “dimensionless constants” will be our focus.

energy density of empty space.

One such constant is the cosmological constant, which measures the

The cosmological constant is just a number which (as far as we know) could have different values consistent with the laws of physics.

likely it is, given the laws of the nature, that the fundamental constants (like the cosmological constant) would fall in a certain range.

One thing that the standard model of physics gives us is a measure of how

We can make certain plausible assumptions about what it would take for life to have evolved. For example, if there were nothing but hydrogen, it is to see how life could have evolved.

hard to see how life could have evolved. If there were no planets, it is hard

Given assumptions such as these, we can look at what the standard model constant has a value which would permit the evolution of life.

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement. (There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

This is the claim that that the cosmological constant having a life-supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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cosmological constant falling in a life-supporting range, given the laws of

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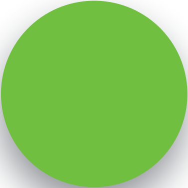
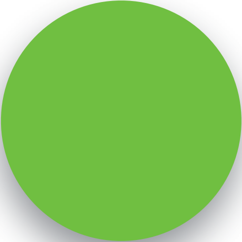
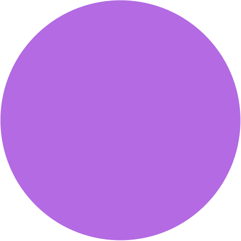
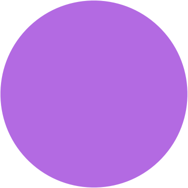
fine-tuning of the universe

Bayes' theorem









cosmological constant falling in a life-supporting range, given the laws of

The claim is that, according to current physics, the probability of the nature, is

$$\frac{1}{10^{120}}$$

Let's now ask how to turn these facts about current physics and probability theory into an argument for the existence of God.

in the urn, so we can consider two different hypotheses about the origins

Much as we considered two different hypotheses about the number of balls of the universe.

hypothesis that it was not.

created by an intelligent designer. Let the **non-design** hypotheses be the

Let the **design hypothesis** be the hypothesis that the universe was

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about the relevant probabilities?

$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$Pr(e \mid \text{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the nonnumbers.

design hypothesis given our evidence is a matter of just plugging in the

$$Pr(\text{non-design}) = 0.5$$

$$Pr(\text{design}) = 0.5$$

$$Pr(e) = 0.25$$

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

consider the odds of winning Powerball one trillion times in a row. Call that a "super to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

It is difficult to think about numbers as large as the denominator of this fraction. But

Powerball."

Now consider the odds of winning a super Powerball one trillion times in a row. Call that a “super duper Powerball.”

The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the odds of the universe being life-permitting by chance.

Now consider the odds of winning a super duper Powerball one trillion times in a row.

This means that if you simply take the physics at face value, and begin by assigning a probability of 0.5 to the non-design hypothesis, you should think that the chances of super duper powerball a trillion times in a row.

the non-design hypothesis being true are vastly lower than the chances of winning a

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1 minus the very small number we were just discussing:

$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.

the argument

objections

3. P(design)=P(non-design)=0.5.

. (1,2,3,4)

4. Bayes' theorem.

2. P

5. P

1. P

8. If the universe was created by an

7. $P(\text{design}) \approx 1$. (5,6)

intelligent designer, then God exists.

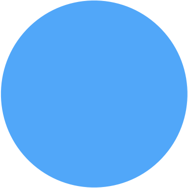
6. Life exists.

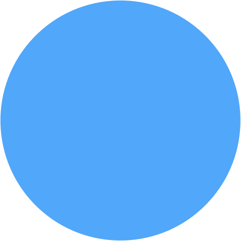
C. $P(\text{God exists}) \approx 1$. (7,8)

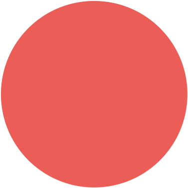
$$(\texttt{life}\,|\,\texttt{design}) = 0.5$$

$$(\texttt{life}|\texttt{non-design}) = \frac{1}{10^{120}}$$

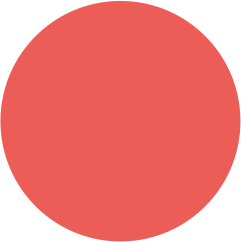
$$(\texttt{design}|\texttt{life}) \approx 1$$







fine-
tuning of the
universe



There would appear to be two reasons why not.

very implausible.

just making a wildly implausible hypothesis just less wildly implausible — but still

First: the hypothesis that Satan controls the weather in South Bend is antecedently extremely implausible. So even if this piece of evidence favors it, it is

$$P(a|b) = \frac{P(a\&b)}{P(b)}$$

To go back to the main lecture, click here.

Derivation of Bayes' theorem

1. $P(a\|b) = \frac{P(a\&b)}{P(b)}$	def. of conditional probability
2. $P(b\|a) = \frac{P(a\&b)}{P(a)}$	def. of conditional probability
3. $P(a\|b) * P(b) = P(a\&b)$	(1), multiplication by =';s
4. $P(a\&b) = P(b\|a) * P(a)$	(2), multiplication by ='s
5. $P(a\|b) * P(b) = P(b\|a) * P(a)$	(3),(4)
C. $P(a\|b) = \frac{P(b\|a) * P(a)}{P(b)}$	(5), division by ='s

Remember our definition of conditional probability:

Using this definition, we can prove Bayes' theorem as follows:

Appendix: Proof of Bayes' theorem